Darksaber Clash

Bob is making a hit new game called Blackhole Wars, which is a 2D platformer with awesome sword mechanics. Since in this game everyone has a Darksaber™ sword, Bob wants to add parry mechanics. First, he needs your help to write an algorithm that can determine whether or not the two swords have intersected.

Problem Statement

You are given N test cases. Each test case will have eight integers separated by spaces, $x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4$. The swords are all line segments. The points (x_1, y_1) and (x_2, y_2) define the location of the first sword while the points (x_3, y_3) and (x_4, y_4) define the location of the second sword. If the swords are intersecting, print the point of intersection. Otherwise, print -1. For this problem, you may assume that the lines will not be colinear and that all four of the given points are unique.

Input

The first line contains an integer N where $1 \leq N \leq 1000$. The next N lines each contain eight space-separated integers $0 \leq x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4 \leq 1000$.

Output

If it exists, print "$x_i \;\; y_i$" without the quotes where (x_i, y_i) is the point of intersection between the first sword and the second sword. Otherwise, print -1. Your answer will be considered correct if its absolute or relative error does not exceed 10^{-6}.

Sample Input

```
3
0 0 5 5 5 0 0 5
0 4 4 4 2 0 2 6
3 3 5 5 1 0 0 1
```

Sample Output

```
2.5 2.5
2 4
-1
```